



Population viability analysis of the endangered copper redhorse *Moxostoma hubbsi*

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ABSTRACT: Studying population dynamics of freshwater fishes facing extinction is becoming increasingly important to better understand how to protect them in our fast-changing environment. In this study, our objective was to better understand copper redhorse *Moxostoma hubbsi* (an endangered fish species only found in Quebec, Canada) population dynamics and the effect of stocking to quantify its risk of extinction. We used stochastic population dynamics models based on current survival and reproduction knowledge of copper redhorse. Although our models suggested that extinction probabilities were low, they show that the population is sensitive to random variability and that it is highly likely for copper redhorse population abundance to become low (<250 individuals), making them vulnerable to genetic impoverishment. Also, the low survival of the younger age-classes made them of greater concern for conservation. Our model shows larvae and fry stocked in the Richelieu River contribute to maintaining the copper redhorse population; however, it also emphasizes the importance of diversifying the current program by stocking 1 yr old individuals to increase population size and help prevent extinction.

KEY WORDS: Conservation · Fish · Stocking · Freshwater · Protection · Modeling

1. INTRODUCTION

Among the many fish found across Canada, copper redhorse *Moxostoma hubbsi* Legendre, 1952 has a global distribution that is limited to a small portion of southwestern Quebec, Canada (Fig. 1). In addition to copper redhorse, 4 other redhorse species are found in the Richelieu River, but have a greater range and are found in many freshwater bodies across North America: greater redhorse *M. valenciennesi* Jordan, 1885; river redhorse *M. carinatum* (Cope, 1870); shorthead redhorse *M. macrolepidotum* (Lesueur, 1817); and silver redhorse *M. anisurum* (Rafinesque, 1820). These 5 redhorse species spawn once a year in similar habitats of the Richelieu River but during distinct time

periods depending on water temperature (Mongeau et al. 1992). In the Richelieu River, copper redhorse is the last to spawn, typically from the last week of June to the first week of July (Committee on the Status of Endangered Wildlife in Canada [COSEWIC] 2014). The Richelieu River is extremely important to copper redhorse because it is, to date, the only river where it is known to spawn (2 known spawning grounds in the river) and where nursery habitats are found. Sexual maturity of copper redhorse in the Richelieu River is also quite late (age 10 yr) compared to greater (9 yr), shorthead (6 yr), and silver (8 yr) redhorses (Mongeau et al. 1992). Only river redhorse reaches sexual maturity as late as copper redhorse. Compared to other redhorse species, copper redhorse has the longest life

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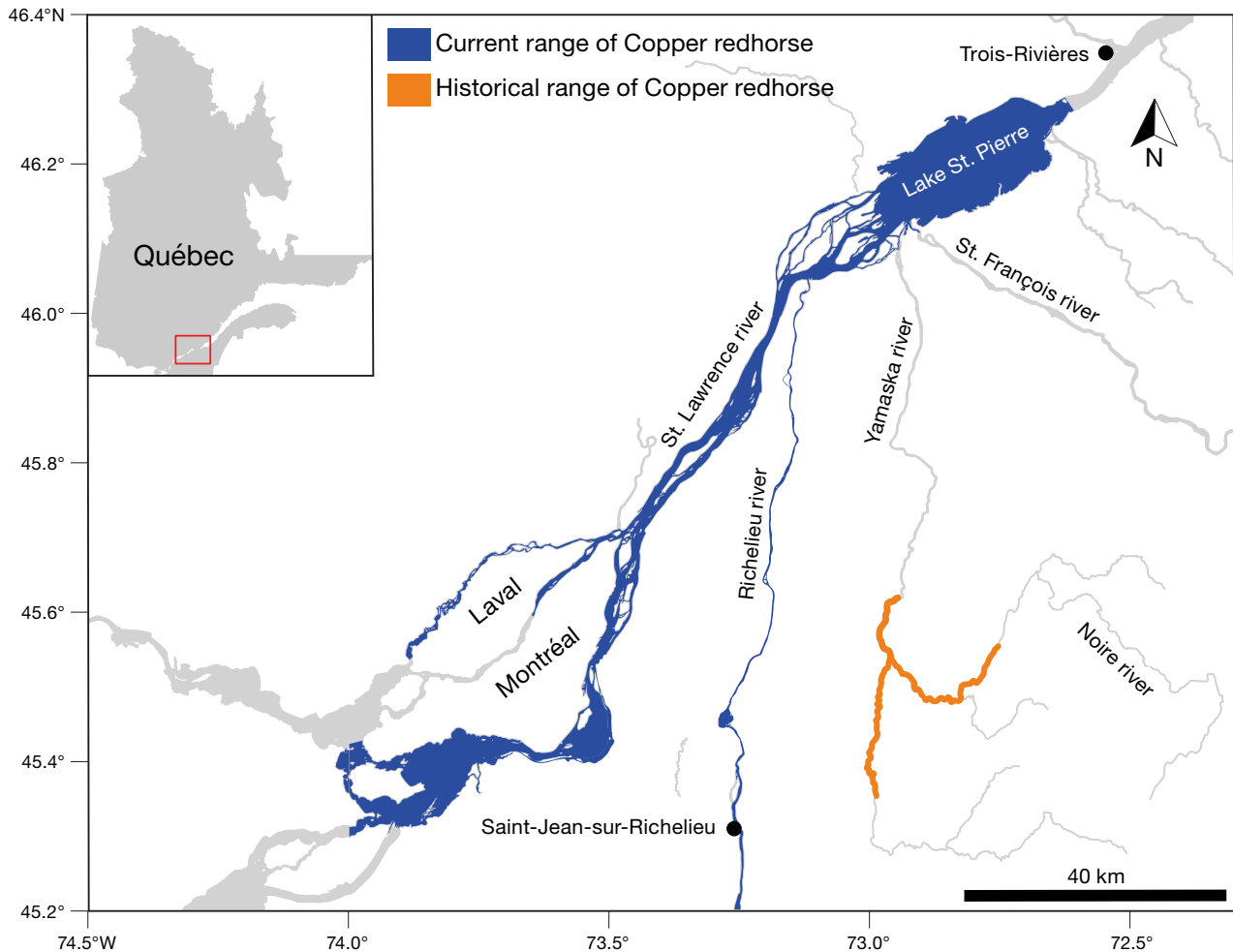


Fig. 1. Current and historical global distribution of copper redhorse in Quebec, Canada; modified from COSEWIC (2014). Copper redhorse were historically found in the Yamaska and Noire river system but are most likely extirpated from this area

expectancy (>30 yr) and produces the most eggs per kilogram of mass (Mongeau et al. 1992).

Freshwater fishes are among the most threatened groups of animals in North America, with nearly 40% of North American freshwater fish species at risk of extinction (Jelks et al. 2008). Threats to freshwater fish populations include habitat fragmentation, degradation, destruction, overexploitation, invasive species, and climate change (Cooke et al. 2005, Dudgeon et al. 2006, Hughes 2021). Copper redhorse *M. hubbsi* has a distribution limited to a small portion of southwestern Quebec, Canada (Fig. 1) and is designated as threatened under Quebec's Act Respecting Threatened or Vulnerable Species (since 1999) and endangered under Canada's Species at Risk Act (since 2007). Like many other freshwater fishes, copper redhorse is thought to be impacted by its proximity to densely populated areas, as its distribution is re-

stricted to waters surrounded mainly by urban or agricultural land in southwestern Quebec (Fig. 1).

Degradation and loss of its habitats caused by habitat fragmentation (dam construction) and by agricultural, urban, recreational, and commercial development are among the greatest threats. Agricultural practices, which are also widespread, especially in the Richelieu River watershed, contribute to the contamination, siltation, and eutrophication of the habitat. Invasive species introduction, e.g. tench *Tinca tinca* and round goby *Neogobius melanostomus*, is also of concern, but their impacts on copper redhorse are largely unknown. Some intrinsic biological characteristics of the species contribute to its vulnerability; it is a specialist, feeding mostly on mollusks, and has a late spawning season (during a period of low water flow which corresponds to peak pesticide application periods in the field and a period of the year where

boaters are very numerous and active; COSEWIC 2014). While the long-term impact of pesticides at the population level is unknown, it is still of concern, as recent studies have shown adverse effects of contaminated river water on the early life stage of copper redhorse (Marchand et al. 2022).

Many conservation measures have been put in place in the last 40 yr to protect copper redhorse across its distribution area. The copper redhorse's status provides the species and its habitat legal protection; it is prohibited to kill, disturb, harass, or capture copper redhorse under the Canadian Species at Risk Act (DFO 2012, COSEWIC 2014). The Vianney-Legendre fish ladder at the Saint-Ours Canal National Historic Site, constructed in 2001, allows fishes, including redhorses, to move upstream in the Richelieu River to reach the Chambly rapids for spawning (COSEWIC 2014). The Pierre-Étienne-Fortin wildlife refuge was also created in 2002 to reduce disturbance of spawners at the Chambly spawning ground and to protect eggs during incubation. Furthermore, multiple changes have been made in sportfishing and commercial regulations since the 1980s (COSEWIC 2014). In addition to these measures, the Quebec government also implemented an assisted reproduction program in 2004 to support the population by increasing the spawning stock because observations indicated low natural recruitment, an aging population, and rarefaction in its distribution area (Mongeau et al. 1992, Vachon 1999, Bernatchez 2004). Each year, male and female copper redhorse are captured, kept temporally in captivity near the Richelieu River for gamete extraction and egg production, and released in the Richelieu River afterwards (Vachon 2010). Using this assisted reproduction program, copper redhorse larvae are stocked in July, and fry (roughly 3 mo old individuals) are stocked in September. Young individuals are stocked so that they can acclimate at an early age to their wild environment, maintaining the population as wild as possible (P. Dumont pers. comm., Nov 16, 2023). An annual young-of-the-year (YOY) survey that began in 1999 in the Richelieu River continues to monitor the natural recruitment of the species and is also used to monitor genetic variation in YOY copper redhorse.

Population dynamics models are widely used in fisheries resource management, e.g. to regulate the exploitation of the resource in a sustainable manner, but can also be geared towards the conservation of endangered species (Jensen 1974, Young & Koops 2014, Gebremedhin et al. 2021). In a conservation perspective, the models are often referred to as a population viability analysis or a vulnerability analy-

sis, where the focus is on determining the probability of extinction of the species under study within a time frame. This type of analysis has become important in conservation, not only to better understand species, but also to assess their conservation status. For example, to determine the status of a species, the E criterion of the IUCN, a criterion that is also used by the COSEWIC in Canada, uses the probability of extinction of a species within the next 5 generations and is often based on population viability analyses (IUCN 2012).

The objective of this study is to provide a population viability analysis for copper redhorse in order to better understand its current situation, extinction probability, trend in population size, and effects of stocking to maximize its long-term survival. In this paper, 3 stochastic population dynamics models with varying degrees of survival and fecundity were developed, extinction probabilities of copper redhorse were calculated, and the effect of different stocking measures were compared to determine the efficacy of the current assisted reproduction program.

2. METHODS

2.1. Data

Our data come in large part from 2 studies on copper redhorse: Mongeau et al. (1986, 1992), who provided catch curve data, i.e. the number of individuals caught in each age class, along with their length and fecundity data. But given the low catch rate of younger individuals by Mongeau et al. (1986, 1992), we supplemented their length data by using lengths obtained from the annual copper redhorse survey and copper redhorse hatchery, where older individuals (age 1+ yr) were occasionally raised (Vachon 2020, 2021b,c). We also estimated survival values of YOY using freshwater fish egg survival data from Dahlberg (1979). This information was used to build the Leslie matrices used in our matrix population models.

2.2. Leslie matrices

We used Leslie matrices (Leslie 1945) to study the population dynamics of copper redhorse. A Leslie matrix characterizes the survival and fecundity of predefined classes of a population and is used to quantify transitions between classes. In the present study, each class represented 1 yr, except for the first year of life, which was split into egg, early life, and fry

groups. The choice to partition the first year into these 3 groups was made to study the effect of stocking and because not accounting for the variation in survival across the different first-year life stages would not be representative of copper redhorse natural history.

Our models were developed using the survival and fecundity of copper redhorse at each age class, as was previously done for the black redhorse *Moxostoma duquesnei* by Young & Koops (2014), who estimated survival by combining a von Bertalanffy growth curve (von Bertalanffy 1938) parameterized with the size-dependent mortality model of Lorenzen (2000). Specifically, survival for each age class was estimated using:

$$s_i = \left[\frac{L_i e^{-k}}{L_{i+1}} \right]^{-\beta} \quad (1)$$

where s_i is survival at age class i , L is the total fish length in millimeters, β is the slope of the catch curve (the number of individuals caught at each age class) for individuals aged ≥ 5 yr, and k is the growth parameter estimated from the von Bertalanffy growth equation. β was estimated using individuals of aged ≥ 5 yr, as the low catch rate of young individuals led to an overly high estimate of survival in younger fish, much higher than what is typically found in species of the same genus, like black redhorse (Young & Koops 2014). Given that length was unavailable for later stages (over 21 yr old individuals), we estimated a linear growth rate for older individuals equal to the growth rate of 18, 19, and 20 yr old individuals.

First year survival was subdivided into 3 groups and estimated independently. For copper redhorse, the egg group lasts roughly 20 d (~5–6 d from spawning to hatching and ~14 d of yolk sac larva), the early life group spans ~60 d (from the end of the egg group to around the time the abundance is measured by the annual survey), and the fry group lasts for the remainder of the first year (from the end of the early life stage to age 1 yr). To estimate survival for these 3 groups, we developed a logarithmic curve of survival with a minimum value of 0.84, which is the mean daily survival of the freshwater fish egg data collected by Dahlberg (1979), and a maximum value of 0.999, which is the daily survival of individuals in the year-1 age class of copper redhorse. In doing so, we were able to multiply together the daily survival from hatching to Day 14, Day 15 to Day 75, and finally from Day 76 to Day 365, which yielded the survival of the 3 groups. For the egg groups, we also multiplied daily survival from hatching to Day 14 by 5.5 because copper redhorse eggs take, on average, 5.5 d to hatch. Yearly sur-

vival curves and daily survival curves for the first year are shown in Fig. S1 in the Supplement at www.int-res.com/articles/suppl/n057p021_supp.pdf. Combined, these 3 groups last 1 yr, and multiplying their survival represents the first year survival of copper redhorse.

Accounting for the variation of fecundity across age classes is essential in the construction of a Leslie matrix. Fecundity was estimated based on the mean length of individuals at each age class measured by Mongeau et al. (1992), who proposed estimating the number of eggs produced by a female (n_{egg}) as:

$$n_{\text{egg}} = 0.00009795024 \times L^{3.1803} - 11367 \quad (2)$$

where L is the total fish length in millimeters; these values were recently corroborated by Vachon (2021a). Copper redhorse length for advanced age classes (22+ yr old) was estimated using a linear growth curve with a slope equal to the mean growth from the 3 previous ages, recognizing that, like survival, their growth is essentially linear at advanced ages. Thus, based on Eq. (1) and available data relating the length of individuals to age, we estimated the fecundity within each age class. For this study, we established that copper redhorse would live a maximum of 30 yr, given that individuals older than this would be extremely rare, and we assumed sexual maturity was reached at 10 yr (COSEWIC 2014), meaning no individual reproduces before this and all individuals reproduce once they reach 10 yr of age.

Regarding reproductive success, we assumed all mature females reproduce because we have no way of estimating this value. We also implemented the suggestions of Kendall et al. (2019) for a post-breeding census system, by accounting for the survival of mature individuals into fertility; i.e. individuals of each age class must survive until reproduction to produce eggs, which is like surviving to the next time step. We multiplied the number of eggs that a female could produce in a time step by the survival of that age class. Due to the absence of data on sex ratio and to consider only the number of egg-laying reproducing females, the number of mature individuals reproducing was also multiplied by 0.5 to account for a 1:1 sex ratio.

The eigenvalue λ of a Leslie matrix can be interpreted as the asymptotic growth rate of the population and, therefore, represents the long-term growth rate of the population. In a population where the class distribution is stable, a population with $\lambda > 1$ is expected to grow, while $\lambda < 1$ means that a population is expected to decline and eventually go extinct. To account for our limited knowledge on copper redhorse, we adjusted survival and fecundity values with

the goal of comparing 3 scenarios. The measured scenario (pessimistic scenario) has a growth rate of $\lambda = 0.957$ and relied on survival and fecundity values derived from the data (Table 1). The endangered scenario has a growth rate of $\lambda = 0.984$ and would qualify copper redhorse as an endangered species under COSEWIC criterion A1 (Table S1). This growth rate was established by following it over 3 generations (75 yr), which would lead to a reduction of 70% of the total number of mature individuals. Given the longevity of this species and the low abundance, a generation time of 25 yr may seem long; however, COSEWIC (2014) considers it to be 25 yr due to the aging of the population. As for the null scenario (optimistic

Table 1. Survival and fecundity values of each age class (years after being a fry) of copper redhorse used for the measured scenario of our stochastic population dynamics models. Values in square brackets define the rows and columns each value takes in a Leslie matrix. The age-30 class has no survival value, as none of the individuals would reach a future age class, and as such, none of them would reproduce, given that reproduction happens right before reaching a new age class, and so they do not have a fecundity value either

Age class	Survival	Fecundity
Egg	0.073 [2,1]	0 [1,1]
Early life	0.023 [3,2]	0 [1,2]
Fry	0.006 [4,3]	0 [1,3]
Age 1	0.609 [5,4]	0 [1,4]
Age 2	0.723 [6,5]	0 [1,5]
Age 3	0.778 [7,6]	0 [1,6]
Age 4	0.812 [8,7]	0 [1,7]
Age 5	0.836 [9,8]	0 [1,8]
Age 6	0.851 [10,9]	0 [1,9]
Age 7	0.863 [11,10]	0 [1,10]
Age 8	0.875 [12,11]	0 [1,11]
Age 9	0.887 [13,12]	0 [1,12]
Age 10	0.893 [14,13]	18 096 [1,13]
Age 11	0.895 [15,14]	21 035 [1,14]
Age 12	0.898 [16,15]	24 114 [1,15]
Age 13	0.902 [17,16]	27 133 [1,16]
Age 14	0.903 [18,17]	29 563 [1,17]
Age 15	0.905 [19,18]	31 934 [1,18]
Age 16	0.908 [20,19]	34 142 [1,19]
Age 17	0.910 [21,20]	35 761 [1,20]
Age 18	0.910 [22,21]	37 062 [1,21]
Age 19	0.910 [23,22]	38 435 [1,22]
Age 20	0.911 [24,23]	39 821 [1,23]
Age 21	0.912 [25,24]	40 936 [1,24]
Age 22	0.911 [26,25]	41 923 [1,25]
Age 23	0.911 [27,26]	42 972 [1,26]
Age 24	0.912 [28,27]	44 075 [1,27]
Age 25	0.912 [29,28]	45 086 [1,28]
Age 26	0.912 [30,29]	46 025 [1,29]
Age 27	0.913 [31,30]	46 914 [1,30]
Age 28	0.913 [32,31]	47 767 [1,31]
Age 29	0.913 [33,32]	48 568 [1,32]
Age 30	—	— [1,33]

scenario), it has a growth rate of $\lambda = 1$, which assumes that the population size remains stable (Table S2). Table 1 presents the measured scenario structured in a summarized format, while Tables S1 & S2 present, respectively, the endangered and null-growth scenarios using the traditional Leslie matrix format to aid with comprehension.

The endangered status of copper redhorse has led to conservation initiatives, such as stocking of larvae and fry, to prevent its extinction (COSEWIC 2014). Therefore, the population dynamics of copper redhorse were also modeled by accounting for annual stocking following actual and theoretical scenarios. The actual stocking scenario is based on the procedure currently used to stock copper redhorse (Vachon 2010), where on average 231 185 larvae and 14 868 fry were stocked annually between 2004 and 2019. A theoretical stocking scenario proposed by Bernatchez (2004) recommended that 500 000 larvae and 15 000 fry be stocked, based primarily on genetic considerations. The proposed stocking quantities of fry and larvae were based on the estimated maximum amount of fry and larvae that could be raised at the Baldwin-Coaticook government fish hatchery that raises copper redhorse fry and larvae in Quebec. These stocking scenarios were included in the measured, endangered, and null-growth population dynamics scenarios, by adding every year current or theoretical stocking numbers of larvae and fry to the population, as if individuals were born into these groups every year.

In addition, we evaluated the effects of stocking larvae, fry, age-1 individuals, and sexually mature (age-10) individuals. For simulation purposes, we established a maximum stocking value of mature individuals after 100 yr at 5000, meaning that we tested how many individuals of these different age groups needed to be stocked yearly to reach 5000 mature individuals after 100 yr. Although the choice of using a threshold of 5000 mature individuals was arbitrary, it allows us to clearly distinguish the differences among stocking individuals from different age groups. We also implemented a 10 yr gap when accounting for stocking mature individuals, as it would take 10 yr before producing the first cohort to stock.

2.3. Population dynamics analyses

Using the Leslie matrices constructed for the different scenarios discussed in Section 2.2, we examined the population dynamics of copper redhorse. To assess population dynamics across an extended period,

we ran simulations for 100 yr, because 5 generations exceed this time limit, as suggested by the IUCN when 5 generation times is >100 yr (COSEWIC 2014).

As the latest estimated number of mature copper redhorses remaining in the wild is estimated to range from a few hundred to a few thousand (COSEWIC 2014), we performed 3 sets of simulations with an initial number of eggs in each scenario to have 500, 1000, and 2000 mature individuals at year 1. The abundances of the other age classes were calculated according to survival probabilities. Therefore, population size was calculated as:

$$n_{\text{age}+1,t+1} = n_{\text{age},t} \times s_{\text{age}} \quad (3)$$

where n_{age} is the number of individuals in a particular age class and s_{age} is the survival for this age class. Given that the time spent at the egg and early life stages are notably shorter than for the other age classes, the first age class in the population dynamics model was calculated to represent the number of fry in year t and was calculated by multiplying the number of eggs laid by the survival probability of these eggs (representing the number of early life individuals), which was then multiplied by the early life group survival probability. Through this calculation, while the first year is subdivided into 3 groups, first year survival is calculated over the same period as all the other age classes.

Small changes in mortality can have drastic effects on recruitment and population size (Houde & Hoyt 1987). Therefore, the effect of random variability influencing rates of survival and fecundity was also considered. Fecundity rates of each age class were assumed to fluctuate yearly based on a factor following a normal distribution with a mean of 0 and a standard deviation of 0.25, allowing fecundity to vary between 0 and twice the mean fecundity value for every given age class. As for survival rates, we assumed it followed a β distribution with an α -shaped parameter of $(s_{\text{age}} \times \beta)/(1 - s_{\text{age}})$, and a β -shaped parameter varying for the 3 first groups, as 5.05, 20, and 75 respectively, and set at 0.75 for all other age classes. In other words, the α and β parameters were defined such that the mean $(\alpha/(\alpha + \beta))$ of the β distribution is s_{age} and the survival values vary between 0 and 1. Using this formulation of parameters and under the assumption that environmental fluctuations can have a much stronger impact on the survival of young individuals than on the survival of older ones, as individuals grow older, the variation in survival will diminish to $\pm 7.5\%$ around the mean for the oldest individuals.

Stochastic models were also used to calculate the probability of extinction of copper redhorse across

the different scenarios. Each stochastic model was run 5000 times to assess the probability of extinction for each scenario. We elected to run each stochastic model 5000 times to make sure the results were stable. Preliminary runs showed that analyses performed with 1000 iterations yields the same results. A population was considered extinct if its abundance reached 0 individuals at any point over the course of the 100 yr. We also calculated the probability of mature copper redhorse abundance being above a certain abundance (the probability of the mature population abundance being >0 individuals each year over the next 100 yr, to the probability of being >1000 individuals over the next 100 yr) across all years, as there are often concerns of genetic diversity loss in small populations (Lippé et al. 2006).

3. RESULTS

Our population dynamics models and viability analysis allowed us to investigate how sensitive the population of copper redhorse is to random fluctuations by accounting for stochasticity and to quantify its risk of extinction. Focusing on the set of simulations carried out with an initial 1000 mature individuals, we found that the different scenarios have major impacts on the abundance of mature individuals and that the abundance of mature individuals can fluctuate greatly. Our models highlight low probabilities of extinctions within 100 yr in all our scenarios (<2%) but strong population decline in our measured scenario and very high likelihood for copper redhorse to fall to low abundance values in all scenarios (Fig. 2).

Our study also evaluated the decrease in population size of copper redhorse after 100 yr by calculating the probability of the population to fall below a specific number of individuals ranging between 0 and 1000 (Fig. 3). All scenarios exhibited high variability, especially for low threshold values. In addition, the median curve highlights that it is quite likely to reach low population abundances. Focusing on this result, in the null scenario considered (the most optimistic scenario for population recovery), over the course of 100 yr, the population would be <250 mature individuals over 30% of the time, while in the measured scenario, this situation is expected to occur over 80% of the time (Fig. 3).

The results for the scenarios with 500, 1000, and 2000 mature individuals at year 1 had similar low extinction rates (<3%). Compared to the simulations based on 1000 mature individuals, the 2000 mature-individual scenario resulted in slightly lower risks of

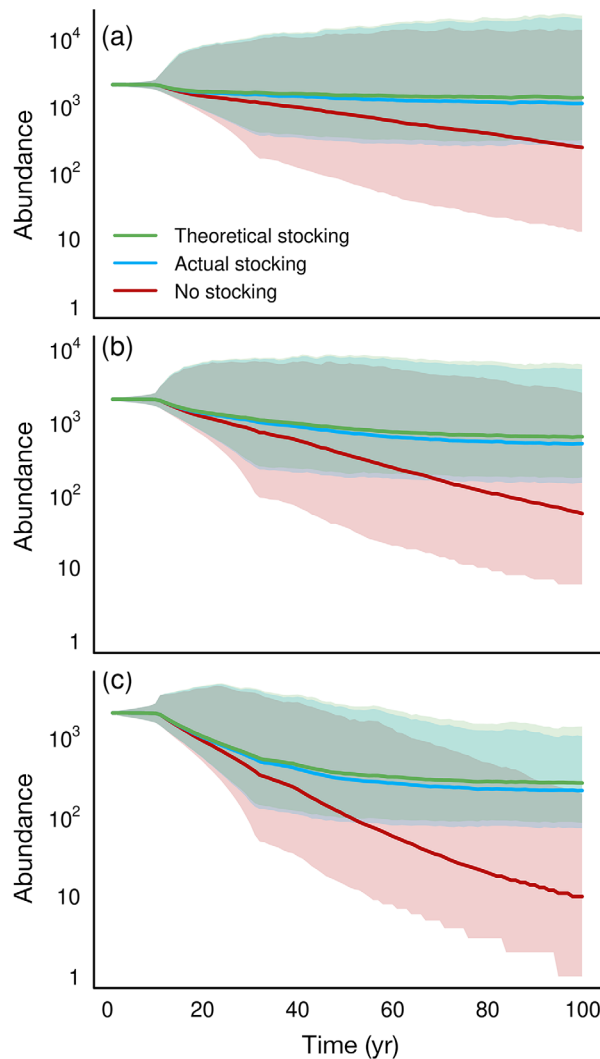


Fig. 2. Median copper redhorse abundance based on 5000 stochastic iterations for the (a) null growth, (b) endangered, and (c) measured scenarios of our stochastic population dynamics models. Colored envelopes: 95% CIs of the abundances; full lines: median abundance for mature individuals

the mature population size falling below 250 individuals (around 20% in the case of the optimistic scenario and around 75% in the case of the measured scenario). Conversely, the 500 mature-individual scenario resulted in higher risk (over 55% in the case of the optimistic scenario and nearly 95% in the case of the measured scenario).

We also examined the effects stocking had on mature population size over 100 yr, to maximize copper redhorses' long-term survival. We found that current stocking is important regardless of which population dynamics scenario was considered (Fig. 2); however, the differences between actual stocking and theoretical stocking was minimal. Our results showed that stock-

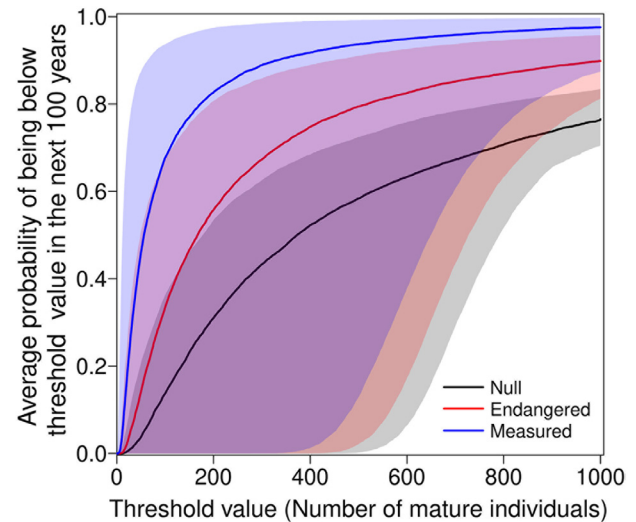


Fig. 3. Probability of copper redhorse to be below a total population threshold within the next 100 yr based on our stochastic population dynamics scenarios. Lines: median of 5000 iterations; colored envelopes: 70% CIs

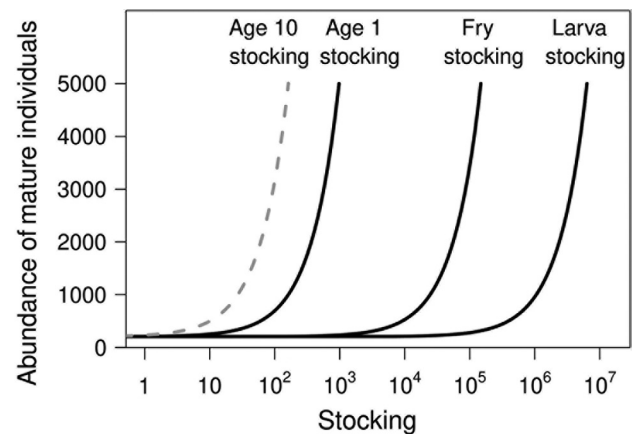


Fig. 4. Effect of increasing stocking of larvae, fry, age-1 individuals, and mature (age-10) individuals on the abundance of mature copper redhorse over 100 yr in our measured scenario on a log₁₀ scale. Age-10 stocking is represented as a dashed line to emphasize that this measure is theoretical and unlikely to be applicable in practice — it is meant to be presented as a comparison to the other stocking options. When applying the same hypothetical stocking measures to the endangered and null-growth scenarios, the results were indistinguishable from the one presented here

ing older individuals had a more pronounced impact on the abundance of mature individuals in the population, regardless of the scenario considered (Fig. 4). Specifically, stocking a few hundred age-1 individuals is equivalent to stocking tens of thousands of fry or millions of larvae (Fig. 4). Stocking fewer, older mature individuals also has a similar effect on the population's abundance according to our model (Fig. 4). Further-

more, the average mature population size after 100 yr was around 800 mature individuals when stocking 500 age-1 individuals, while it was estimated as ~240 individuals if theoretical stocking was to be used (Fig. S2).

4. DISCUSSION

4.1. Population dynamics

These are the first population dynamics models for copper redhorse. Unlike the projections of Bernatchez (2004), which primarily addressed genetic diversity concerns, effective population size, and recommendations on stocking, our study adds a new perspective to copper redhorse conservation. Specifically, we used Leslie matrices to address the survival and fecundity at each age class based on empirical data adapted to the species-specific life history characteristics of copper redhorse and included stochasticity in our models. Our models suggest that current stocking strategies used to conserve copper redhorse population could benefit from being adapted to our results, as the population is likely to decline substantially in the next 100 yr (Fig. 2c), even if extinction probabilities within 100 yr remains low (<2%).

We also found that the stocking rate proposed by Bernatchez (2004; our theoretical stocking) had only a slightly more positive impact than the current actual stocking. Some recent observations in the assisted reproduction program suggest that mature copper redhorse numbers are increasing and that the population is in the process of renewal in the last few years (Vachon 2020, 2021a,b,c), and therefore future analysis including more recent data may shed additional light on current population trends. These recent observations remain consistent with the stochasticity in our models that indicates important (positive and negative) fluctuations in abundance of mature individuals through time, highlighting the important differences across iterations. From a conservation perspective, these fluctuations are concerning, because it suggests that the copper redhorse population is sensitive to environmental fluctuations affecting its survival and fecundity. For example, extreme hydrologic and thermal conditions during incubation and early larval development could lead to lower survival during these stages. Similarly, more intense and frequent precipitation can lead to significant sediment inputs in the Richelieu River, increasing turbidity and contaminants in spawning grounds and nursery habitats. In general, the increase in turbidity in watercourses is detrimental to the develop-

ment of aquatic grass beds (essential for YOY foraging) and thus food resources (Vachon 1999, 2003). Copper redhorse may also be more sensitive than other redhorse species in their first year because they are smaller in size at this life stage (COSEWIC 2014). Furthermore, a general lack of resources causing malnutrition could also impact the fecundity (both quality and quantity of eggs produced) of females (Izquierdo et al. 2001). The opposite is also true — abundant food resources could lead to higher-than-average survival and fecundity. Any events (natural or anthropogenic) that have even a small impact on the survival or fecundity of copper redhorse can have major impacts on its population dynamics and its ability to be maintained through time (Fig. 3). Similarly, Young & Koops (2014) found that black redhorse in Canada was particularly sensitive to changes in the survival of young and immature adult individuals. High variability in population size across 100 yr was also found in a population dynamics study of robust redhorse *Moxostoma robustum* (Jennings et al. 2000).

Bernatchez (2004) recommended stocking a large number of fry, but, due to logistical issues, added that young larvae may also need to be stocked, as the number of fry needed exceeds the capacity of the hatchery. Our results suggest that YOY survival is particularly low, a life stage essential in the conservation of the species to prevent its extinction. Given that their natural survival in the wild cannot be increased easily, the only way to improve the populations' survival is by stocking older individuals, allowing them to live their most vulnerable life stages in captivity. This means that, instead of stocking younger larvae, focus should be placed on stocking slightly older individuals, as they have the best chance of survival; however, given the uncertainty surrounding copper redhorse survival, the stocking of age-1 individuals should not replace the current stocking measures, but rather be added to them.

Our results show that the impact of stocking larvae and fry on the abundance of mature individuals is only evident when many millions of larvae or many tens of thousands of fry are stocked. In this respect, stocking a few hundred age-1 individuals is likely to have substantially greater impact on the copper redhorse population. While according to our models, stocking much older individuals, such as mature individuals (age-10 individuals), would most likely have an even greater impact on short-term population abundance, this solution has many drawbacks and is not a recommended solution. They may lose their spawning homing capacity, have diminished survival, fecundity, and likelihood of reproducing in the wild

and may develop maladapted phenotypes and behavior to their natural environment (Gro Veia Salvanes & Braithwaite 2006, Roberts et al. 2011, Stringwell et al. 2014). Furthermore, rearing enough fish to reach age 10 yr would also be complicated from a logistical standpoint. Holding ponds would have to be quite large to contain all individuals over 10 yr and feeding them would also require a lot of resources. The expertise and know-how required to rear individuals to such a late stage would also need to be developed and is not currently underway (Lamothe et al. 2019).

Several reintroduction programs for catostomids exist in the US, and in many cases, slightly older individuals are stocked (Day et al. 2017). For example, razorback sucker *Xyrauchen texanus* (C. C. Abbott, 1860) stocking efforts have largely been unsuccessful; however, Schooley & Marsh (2007) suggested that, had earlier stocking programs used today's knowledge of size-based release survival and held fish until they reached 350 mm total length (TL), we would have observed more than a 100-fold increase in survivors today. In the case of June sucker *Chasmistes liorus* Jordan, 1878 in Lake Utah, refuge populations were established in reservoirs and the optimal TL for stocking was determined to be 350 mm, once sexual maturity was reached (Day et al. 2017). Fonken et al. (2022) corroborated that stocking slightly older and bigger individuals had a positive effect on the population, concluding that the survival of stocked June sucker was positively correlated with size and that stocking age-2 individuals was more cost-effective than stocking age-1 individuals. Therefore, given our results (Fig. 4) and the other species benefiting from this method, stocking age-1 fish is likely a good compromise between the practicality of rearing individuals over a longer time span and having fish that can adapt and survive in a natural environment and later produce offspring in the wild.

4.2. Conservation implications

Copper redhorse was assessed as endangered by COSEWIC (2014) based on the B1ab(i,iii,iv,v) + 2ab(i,iii,iv,v) criteria, as its occurrence is under 5000 km² and index of area of occupancy is estimated to be under 500 km², it exists in <5 locations, and decline in distribution, quality of habitat, and number of mature individuals was observed and inferred. The species could also be endangered under the C criterion if there are <2500 mature individuals and the mature population is predicted to decline by >20% in the next 3 generation times, in this case 75 yr. Our

modeled stochastic scenario suggests a potential decline of >80%, which would confirm the copper redhorse status as endangered, i.e. facing imminent extirpation or extinction, highlighting the importance of taking immediate actions to protect it and its habitat. Given the recent encouraging observations, it will be important to gather a time series of mature population size to verify the validity of our population dynamics models and if our model accurately predicts copper redhorse population size (Vachon 2021a,b,c). A population decline is not only a direct problem for the population. Since the current stocking method relies on the capture of mature individuals, this means that as time progresses and the mature population size diminishes, catching mature individuals to use in the assisted reproduction program will become more difficult, limiting even further the effect that stocking can have on the population (COSEWIC 2014).

Although human intervention, e.g. stocking, can have a positive impact on population abundance by preventing the strong decline seen in our measured scenario, stocking should remain a preventive measure as, ultimately, copper redhorse should persist without human intervention. This may be achieved by protecting more of its key habitat and increasing our understanding of what is causing low survival and recruitment, as is defined in the copper redhorse recovery strategy (DFO 2012, Équipe de rétablissement du chevalier cuivré du Québec 2012).

An elasticity analysis is often used in population viability and with matrix population models; however, we did not include this analysis in our study. Elasticity analysis is known to be impacted by the structure of the transition matrix (Enright et al. 1995). In our study, we built our Leslie matrix for it to include both life-stage and age classes for each year for up to 30 yr. This was a decision taken to best represent the biology of copper redhorse, given the current knowledge on the species and to include stocking in our model, as explained in Section 2.2. In doing so, we used different approaches to infer information for classes where information was available. As such, performing an elasticity analysis that would single out one or a few classes as important to focus on for acute conservation measures may lead to wrongful interpretation that may lead to ineffective and potentially costly measures. That being said, for completeness, we have included the results of an elasticity analysis performed on the Leslie matrix (Fig. S3), which shows that the survival of younger individuals importantly affects copper redhorse population dynamics. For example, the span of time associated to the first 3 classes (egg, early life, and fry) are different from each other and from the

other age-based classes. This particularity of the transition matrix may artificially give disproportionate importance to the life-stage classes of the first year and artificially skew the interpretation of the results.

4.3. Limitations of this study

Our models and analyses are simplistic because they do not account specifically for many important factors dictating the population dynamics of copper redhorse, such as predation and habitat degradation caused by increasing human activities in its distribution area. This is particularly important because the habitat of copper redhorse, and more broadly the Richelieu River, is commonly used for leisure activities (e.g. boating, water-skiing) and has been under important climatic and pollution pressures (e.g. flood, wastewater discharged, high turbidity, intensive agricultural practices), which may have unknown detrimental impacts on copper redhorse. Furthermore, it was also necessary to make certain assumptions when calculating demographic rates and stochasticity, as these values are not available. For example, the values for YOY redhorse survival derived from Dahlberg (1979) were for few freshwater fish in relatively undisturbed environments and may be overestimated.

We also assumed that the survival of stocked young would be the same as the one observed in wild young; however, this is unlikely to be true, as they are less adapted to the wild conditions found in the Richelieu River. We do not have these data because stocked individuals are too small to be tagged or marked without accruing mortality. Finally, we also assumed that no individual <10 yr of age is mature, and all individuals aged ≥ 10 yr are mature, since no data on the distribution of maturity among age groups is currently available. Nonetheless, our results account for the general mechanisms that affect the demography of copper redhorse (survival and fecundity) and the stochastic nature of the ecosystems in which they live and give new insights into the possible outcomes for the population in the next 100 yr. Our analysis also does not account for genetic variation, which is essential for evaluating at-risk populations, as genetic diversity loss is often a concern for small aging populations such as in copper redhorse.

5. CONCLUSION

Our study emphasizes the precarious state of the copper redhorse population, particularly when we

consider the population's behavior in a stochastic ecosystem. While our models suggest low extinction probabilities, they also suggest that the abundance of mature copper redhorse is often extremely low compared to historical and prehistorical abundances (COSEWIC 2014). When looking at the survival of each age class, we found that the low survival of the younger age classes makes them of greater concern for conservation. Current stocking measures in place in the Richelieu River might benefit from the diversification of stocking strategies by stocking older age-1 individuals to improve recovery of the copper redhorse population, and this measure should be further assessed.

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